Equations of Motion

For a moving object, the relations among its velocity, displacement, time and acceleration can be represented by equations. These equations are called equations of motion.

- For motion a straight line with constant acceleration a

  (i) $v = u + at$ (ii) $s = ut + \frac{1}{2}at^2$

  (iii) $v^2 = u^2 + 2as$

- Equation of motion under gravity, (downward direction)

  (i) $v = u + gt$ (ii) $h = ut + \frac{1}{2}gt^2$

  (iii) $v^2 = u^2 + 2gh$

Where, u = initial velocity

v = final velocity

g = gravitational acceleration

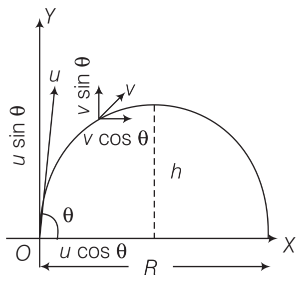
If an object is thrown vertically upward away from the earth, then we shall substitute $-g$ in place of g in the above equations. The value of g is $9.8\ \text{m/s}^2$.

➡ **Note** If a body starts from rest and moves with uniform acceleration, then distance covered by the body in t sec is proportional to t^2(i.e. $s \propto t^2$).

Projectile Motion

When a body moves under an acceleration whose direction is different from the direction of the initial velocity, then both the magnitude and direction of its velocity changes with time. Hence, the body moves on a curved path in a plane. This type of motion is called projectile motion.

- A bullet fired from a rifle and a body dropped from the window of a moving train shows the projectile motion.



- Flight time of projectile, $T = \dfrac{2u\sin\theta}{g}$
- Height of projectile, $h = \dfrac{u^2\sin^2\theta}{2g}$
- Range of projectile, $R = \dfrac{u^2\sin 2\theta}{g}$.
- We drop down a ball from a roof and at the same time throw another ball in a horizontal direction, then both the balls would strike the earth simultaneously at different places.
- The horizontal range is the same when the body is projected at θ and $(90° - \theta)$.
- The horizontal range of a projectile is maximum when angle of projection is 45°.
- When two balls of different masses are projected horizontally they will reach ground at same time.

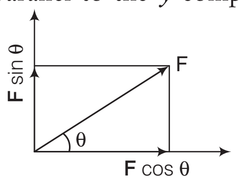
FORCE

Force is a push or pull which produces or tends to produce a change in uniform motion of body; stops or tends to stop a body which is in motion. Its unit is Newton.

- It is a vector quantity and the straight line along which a force is directed is called the line of action of the force.
- The rotational effect of a force on a body about an axis of rotation is described in terms of moment of force.
- CGS unit of force is dyne i.e. 1 dyne $= 10^{-5}$N.

Force is of two types

(i) **Component of Force** Forces acting at same angle from the coordinate axes can be resolved into mutually perpendicular forces called components. The component of a force parallel to the X-axis is called x-component, parallel to the y-component and so on.



Here,
- x-component of force **F** is $\mathbf{F}\cos\theta$.
- y-component of force **F** is $\mathbf{F}\sin\theta$.

(ii) **Contact and Non-contact Force** The forces which act on bodies when they are in physical contact are called **contact forces**. e.g. muscular force, frictional force, etc.

A force which can be exerted by an object on another object even from a distance (without physical contact with each other) is called a **non-contact force**. e.g. magnetic force, electrostatic force and gravitational force.

NEWTON'S LAWS OF MOTION

Newton has given three laws of motion as below:

First Law

It states that every body continues in its state of rest or in uniform motion in a straight line, unless it is compelled by an external force to change that state.

First law is also called **law of Galileo** or **law of inertia.**

- The property of bodies by virtue of which they oppose only change in their present state is called **inertia**.
- Mass is a measure of the inertia of a body.

Here some practical examples are as follows in which first law is used.

- Athlete runs some distance, before taking a long jump.
- A ball thrown upward in a train moving with uniform velocity returns to the hand of thrower.
- The mud from the wheels of a moving vehicle flies-off tangentially.
- When we shake the branch of a mango tree, the mangoes fall down.
- A man jumping from a moving train may fall down.
- If a cloth placed under a book is given a sudden pull, it comes out without disturbing the book.
- When a horse suddenly starts moving, the rider falls backward.
- When a running horse suddenly stops, the rider falls forwards.
- We hit a carpet with a stick to remove the dust.

Second Law

The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts. According to second law

$$F \propto \frac{dp}{dt} \text{ or } F = k\frac{dp}{dt}$$

where, k is constant of proportionality

$$F = k\frac{d}{dt}(mv) = kma$$

for simplicity let $k = 1$.

$$\Rightarrow \quad F = ma$$

$$1 \text{ newton } = 10^5 \text{ dyne}$$

- Newton's second law gives the magnitude of force.
- Newton's first law is contained in the second law.

Here some practical examples are as follows in which second law is used.

- China wires are wrapped in straw or paper before packing.
- A person falling on pucca floor (or frozen ice) is likely to receive more injuries than one falling on kuccha floor (loose earth).
- While catching a ball, a cricket player lowers his hands to save himself from getting hurt.
- Bogies of the trains are provided with buffers to avoid severe jerks during shunting of trains.

Third Law

According to this law, to every action there is an equal and opposite reaction. 'Action and reaction always act on the different bodies.' Here some practical examples are as follows in which third law is used

- In order to walk, we press the ground in backward direction with our feet. As a result, the ground pushes us in forward direction.
- It is difficult to drive a nail into a wooden block without holding the block.
- A jet plane moves on the principle of Newton's third law of motion. As exhaust gases come out from the nozzle at a greater speed, the reaction of the same moves the plane forwards.

Validity of Newton's Laws of Motion

Newton's laws are valid in classical physics only under certain conditions

- In general, the distances which one work must be much greater than the size of atoms and molecules. Else, quantum mechanics must be used in place of classical mechanics.
- In general, the speeds which one work with must be much less than the speed of light. Else, relativistic mechanics must be used in place of classical mechanics.
- Also, this is valid with respect to inertial frame of reference only.

Linear Momentum

- The linear momentum of a body is defined as, the product of its mass and its velocity i.e.
  Momentum = Mass × Velocity, $p = m \times v$
- It's direction is the same as the direction of velocity of the body. It is a vector quantity. It's SI unit is kg-m/s.
- Concept of momentum was introduced by Newton.
- A heavier body has a larger linear momentum than a lighter body moving with the same velocity. i.e. In the absence of external forces, the total momentum of the system is conserved.

  i.e. $m_1v_1 = m_2v_2$ [For single body]

  and $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ [For two bodies]

Applications of Conservation of Momentum

When a bullet is fired from a gun, the gun recoils or gives a sharp pull in backward direction. While firing a bullet, the gun must be held tight to the shoulder.

- When a man jump from a boat to the shore, the boat slightly moves away from the shore.
- Rocket works on the principle of conservation of momentum.
- If someone left on a frictionless floor desires to get out of it, he can do so by blowing air out of his mouth.

Impulse

- An impulse is a large force acting on a body for a short time to produce a finite change in momentum.
- Impulse = Force × Time interval i.e. $I = F \times \Delta t$
  It's SI unit is N-s or kg-m/s

Examples of momentum and impulse are as follows:

- In catching a ball, a player by drawing his hands backward increases the time of contact.
- An athlete is advised to come to stop slowly after finishing a fast race.
- The layer of bricks is broken by player.
- Vehicles like cars, buses, and scooters, are provided with shockers.
- Bogies of the trains are provided with buffers, because buffers increase the time duration of jerks during shunting. This reduces the force with which bogies push or pull each other and severe jerks are avoided.

FRICTION

The opposing force which comes into play when a body moves or tries to move the surface of another body is known as friction.

- The maximum value of the force of friction which comes into play before a body just begins to slide over the surface of another body is called **limiting friction**.
- The force of friction that comes into play between the surfaces of two bodies before the body actually starts moving is called **static friction**.
- The force of friction that opposes relative motion between two surfaces in contact is called **kinetic or sliding friction**.
- The frictional force developed, when a body rolls over a surface, is known as the **rolling friction**.
- Static friction is a self adjusting force and it adjusts itself, so that it became equal to the applied force.

Advantages of Friction

- Due to friction, we are able to move on the surface of the earth.
- The fibres of thread are held together due to force of friction.
- The brakes applied in automobiles work only due to friction.
- Sledges are used in Arctic region as friction is very low on the surface of ice.

Disadvantages of Friction

- A lot of energy is wasted in the form of heat due to friction that causes wear and tear of the moving parts.
- Due to friction, speed of automobiles cannot be increased beyond a certain limit.

Laws of Friction

Friction acts in a direction opposite to the direction of motion of the objects. It depends upon the nature of the two surfaces in contact.

- Friction is independent of the area of contact of the two surfaces. Rolling friction is less than sliding friction.
- Force of friction (F) is directly proportional to the normal reaction (R).

i.e. $F \propto R$

$F = \mu R$

where, μ = coefficient of friction.

Also, $\mu = \tan\alpha$, where α is the angle of friction.

Angle of repose = Angle of friction i.e. $\theta = \alpha$

Methods for Reducing Friction

- By using lubricants e.g. grease, oil, etc.
- By using ball or roller bearings, which changes sliding into rolling.
- By using soap solution.
- By using powder.
- Friction due to air is considerably reduced by streamling the shape of the body moving at high speed in air. e.g. aircraft, jet-planes, fast cars are streamlined.
- The tyres are provided with treads which increases the friction between the tyres and the road.

➡ **Note Pulling is easier than pushing**
While pushing, there is one component of force that acting downward on the object adds to the weight of the body hence there is more friction opposing the effort. Whereas on pulling, the vertical component of force is against the weight of the body and hence there is less overall friction. So, it is easy to pull than push.

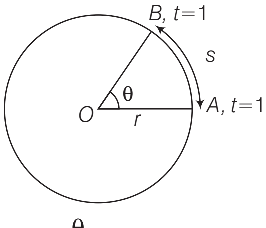
CIRCULAR MOTION

When an object moves along a circular path, then its motion is called circular motion as motion of top, etc. In circular motion force is always at right angles to the displacement therefore no work is done by the force on the particle.

- When an object moves along a circular path with uniform speed, its motion is called **uniform circular motion.**
- Circular motion is accelerated even, if the speed of the body is constant. The motion of a satellite is accelerated motion.

Angular Velocity

Angular velocity is the rate at which angle swept by the radius at the centre changes with time. Its unit is rad/s.



- Angular velocity, $\omega = \dfrac{\theta}{t} \Rightarrow s = v \times t$ and $\omega = \dfrac{\theta}{t}$,

$$\theta = \frac{s}{r}$$

So, $v = \omega \times r$

as v = linear speed,

ω = angular velocity,

r = radius of the circular path.

Thus, linear velocity = Angular velocity × Radius of circular path.

The angular velocity of revolution of a planet around the sun in an elliptical orbit increases. When the planet came closer to sun and *vice-versa*.

Angular Acceleration

- The rate of change of angular velocity is defined as angular acceleration.
- Angular acceleration, $\alpha = \frac{\omega}{t}$. Its unit is rad/$s^2$ or revolution/sec^2.
- Relation between angular acceleration and linear acceleration, $a = \alpha \times r$

Centripetal Acceleration

The acceleration is directed towards the centre and is given by $a = \frac{v^2}{r}$, where v is the speed and r is the radius. This is called centripetal acceleration.

Centripetal Force

A body performing circular motion is acted upon by a force which is always directed towards the centre of the circle. This is called **centripetal force**. Work done by centripetal force is always zero.

- If a body of mass m is moving on a circular path of radius R with uniform velocity v, then the required centripetal force, $F = \frac{mv^2}{R} = m\omega^2 R$
- Any of the forces found in nature such as frictional force, gravitational force, electrical force, magnetic force may act as a centripetal force.
- Cyclist bends his body towards the centre on a turn while turning to obtain the required centripetal force.
- Generally, in rain the scooter slips off the turning of a road because the friction between tyre and road is reduced. Due to this necessary centripetal force is not provided.
- Vehicles can move on circular path safely, if friction force ≥ required centripetal force.
- Roads are banked at turns to provide the required centripetal force for taking a turn.
- Work done by centripetal force is always zero.

Centrifugal Force

There are certain situations in which we feel that a body is acted upon by a force, but actually there is no force on the body. Such an apparent force is called a pseudo force.

- Centrifugal force is such a pseudo force. It is equal and opposite to centripetal force. When a person stand on a rotating platform, then he feels the centrifugal force.
- Cream separator and centrifugal drier work on the principle of centrifugal force.

➠ **Note** Centrifugal is a device to separate the particles of different masses present in liquid.

WORK

Work is said to be done, if a force acting on a body is able to actually move it through some distance in the direction of the force.

- Work = *Force* × *Distance*,
  i.e. $W = F \cdot s$
- If the force F is making an angle θ with the direction of displacement of the body, then the work done is $W = Fs\cos\theta$
- It's SI unit is joule (J).
- CGS unit of work is erg i.e. 1 J = 10^7 ergs

Types of Work

Work can be of following types

1. **Positive Work Done** Positive work means that force is parallel to displacement.
   Examples
   - When a body falls freely under gravitational pull.
   - When a horse pulls a cart on a level road.
2. **Negative Work Done** Negative work means that force is opposite to displacement.
   Examples
   - When a body is made to slide over a rough surface or when a positive charge is moved towards another positive charge.
3. **Zero Work Done** If the force is perpendicular to the displacement and if either the force or the displacement is zero, work done will be zero.
   Examples
   - When a coolie travels on a platform with a load on his head.
   - When a body is moved along a circular path. When a person does not move from his position but he may be holding any amount of heavy load.
   - If the force is perpendicular to the displacement and if either the force or the displacement is zero, work done is zero.

Conservative and Non-conservative Forces

- A force is said to be **conservative**, if the work done by the force in moving a body depends only upon the initial and final positions of the body and is independent of the path followed between the initial and final position, e.g. gravitational force, electrostatic force and magnetic force, etc.
- A force is said to be **non-conservative**, if the work done by the force or against the force, in moving a body from one position to another, depends upon the path followed between the two positions. e.g. friction force, etc.

ENERGY

The energy of an object is its capacity for doing work. It is a scalar quantity and its unit is joule (J).

- Energy can exist in different forms such as mechanical energy, heat energy, sound energy, light energy, chemical energy, etc.
- Mechanical energy is of two types i.e. kinetic energy and potential energy.

1. Kinetic Energy

The energy possessed by a body by virtue of its motion is called its kinetic energy.

- If a body of mass m is moving with velocity v, then kinetic energy $= \frac{1}{2}mv^2 = \frac{p^2}{2m}$
- When momentum is doubled kinetic energy becomes four times.
- If a body is moving in horizontal circle then its kinetic energy is same at all points, but if it is moving in vertical circle, then the kinetic energy is different at different points.

Applications of kinetic energy are given below

(*i*) Kinetic energy of air is used to run wind mills.
(*ii*) Kinetic energy of running water is used to run the water mills.
(*iii*) A bullet fired from a gun can pierce a target due to its kinetic energy.

2. Potential Energy

It is the energy possessed by a body by virtue of its position.

- Suppose a body is raised to a height h above the surface of the earth, then potential energy of body $= mgh$
- Also, here potential energy = work done $= mgh$
- When the body is projected upwards, then its kinetic energy goes on changing to potential energy and *vice-versa*.

Applications of potential energy are given below

(*i*) The potential energy of the wound spring of a clock is used to drive the hands of the clock.
(*ii*) Due to potential energy of the stretched bow, the arrow goes forward with a large velocity on releasing the bow.
(*iii*) The potential energy of water in dams is used to run turbines in order to produce electrical energy using the generators.

Principle of Conservation of Energy

According to the law of conservation of energy it can neither be created nor be destroyed but can only be transformed from one form to another.

- The total amount of energy in the universe remains constant. e.g. when an object falls, its potential energy decreases but at the same time kinetic energy increases by an equal amount.

Transformation of Energy

- In a heat engine, heat energy changes into mechanical energy. In the sun, mass changes into radiant energy.
- In an electrical heater, the electric energy is converted into heat energy.
- In an electric bulb, the electric energy is converted into light energy in burning coal, oil, etc. The chemical energy changes to heat energy.
- When we rub our hands, heat is generated. Here, mechanical energy of muscles is converted into heat energy.
- In an electromagnet the electric energy is converted into magnetic energy.

Einstein's Mass-Energy Equivalence

In 1905, Einstein proved the relation between mass and energy. $E = mc^2$, where c is the velocity of light.

$E = mc^2$ leads to verification of the two laws, law of conservation of mass and law of conservation of energy.

POWER

The time rate of doing work is called **power.**

If W is work done in time t then power, $P = \frac{W}{t}$.

For a constant force F, the

Power $(P) = F \cdot v \Rightarrow P = Fv\cos\theta$

where v = velocity of object

θ = angle between F and v

- Power is a scalar quantity with SI unit watt (W) or joule/s.
- Some other units of power are 1 watt hour = 3600 J

1 kilowatt hour $= 3.6 \times 10^6$ J

1 HP (Horse Power) = 746 W